

UBIQUITY

SECURE REMOTE ACCESS FOR INDUSTRIAL AUTOMATION DEVICES



@ASEM

A ROCKWELL AUTOMATION COMPANY

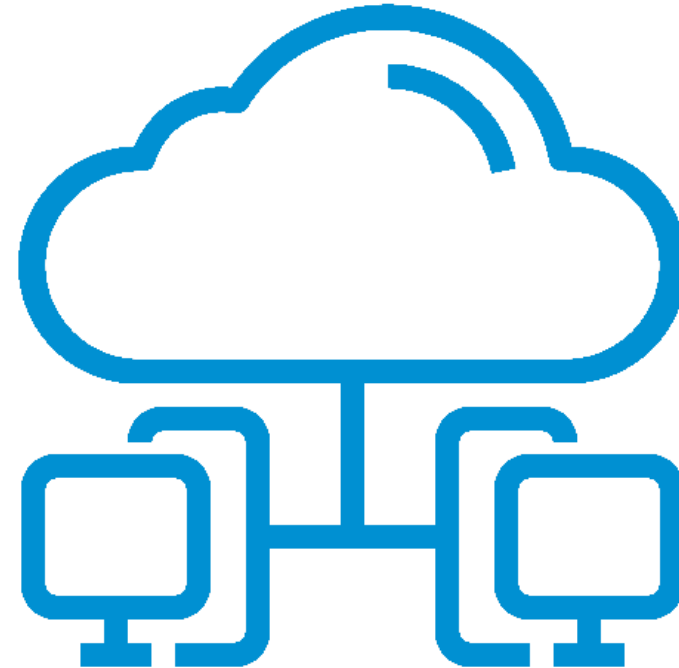
v25.00 – December 25

SUMMARY



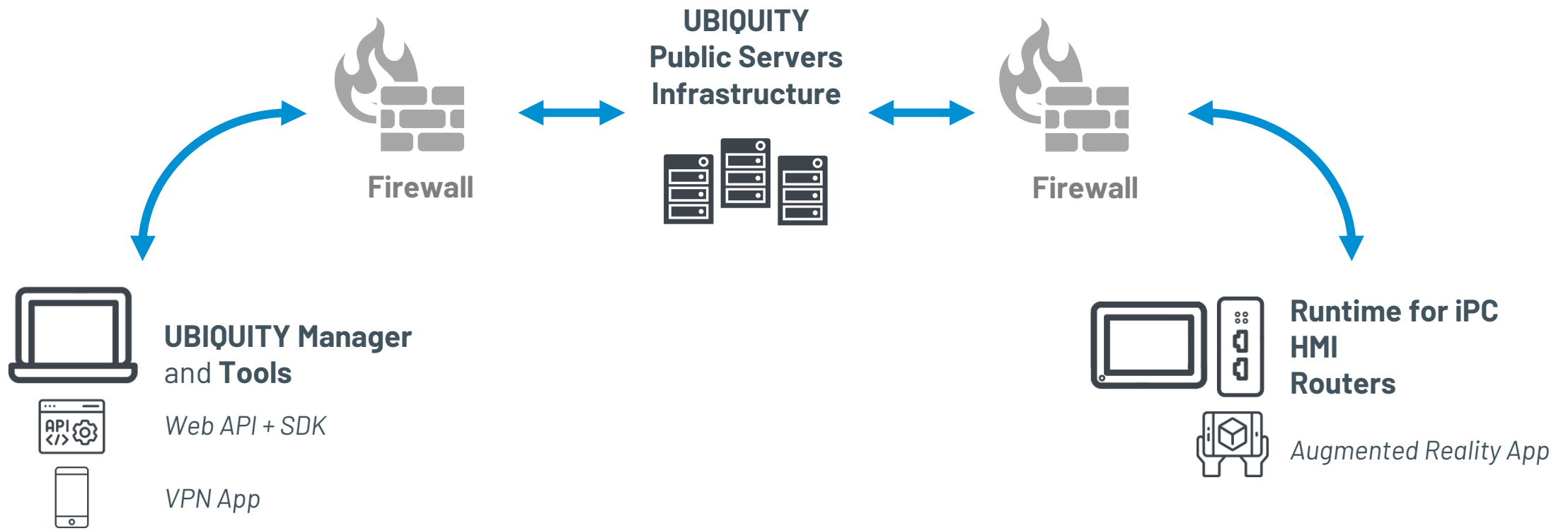
UBIQUITY OVERVIEW PUBLISHED: Copyright ©2025 ASEM a Rockwell Automation Company	RUNTIME FEATURES PUBLISHED: Copyright ©2025 ASEM a Rockwell Automation Company	ROUTER FEATURES PUBLISHED: Copyright ©2025 ASEM a Rockwell Automation Company
CONNECTIVITY SERVICES PUBLISHED: Copyright ©2025 ASEM a Rockwell Automation Company	LICENSE MODEL PUBLISHED: Copyright ©2025 ASEM a Rockwell Automation Company	PRIVATE SERVER PUBLISHED: Copyright ©2025 ASEM a Rockwell Automation Company
UBIQUITY VS FT REMOTE ACCESS PUBLISHED: Copyright ©2025 ASEM a Rockwell Automation Company	UBIQUITY VS UBIQUITY X PUBLISHED: Copyright ©2025 ASEM a Rockwell Automation Company	SECURITY ARCHITECTURE PUBLISHED: Copyright ©2025 ASEM a Rockwell Automation Company

UBIQUITY OVERVIEW



OVERVIEW

- UBIQUITY Manager and UBIQUITY Manager Tools
- UBIQUITY Public Server Infrastructure
- Runtime, HMI and Routers



OVERVIEW: UBIQUITY MANAGER

Ubiquity

+

← → ↻

ubiquity.asem.it/controlcenter/

🔍 🗨 🧑 Guest ⋮

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UBIQUITY

🔍 🧑 Luca Bognini ⌵

Domain View

ASEM Demo: 3 📶

Explorer

Domain view

Devices view

Map view

Audit

Settings

Tools

Log

Search

⌵

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🔔 🔴

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⋮

ASEM Demo

ASEM Verona

Alessio

HMI30 - Alessio

Demo BOX

IT4-Alessandro

Rm21_New

Meeting Room

Demo IIoT Cloud Connector HT700 + Omron Fins Ser&Eth (10.60.0.60)

Ubiquity RK10 + S7-300 MPI/TCP (10.90.0.90)

AUSTRIA

pro-SCADA

intern

HMI30 - Alessio Information

Status

Online

Device IP Addresses

10.8.0.1 📶

172.19.17.40

Public IP Addresses

93.62.67.178

UBIQUITY Runtime Version

13.4.13

Firmware Version

--

Update Status

Up-To-Date

Update Runtime

Operating System

Windows Embedded Compact 7.0

CPU Architecture

ARM

System Family

Generic device

System Capabilities

OVERVIEW: PUBLIC SERVER INFRASTRUCTURE



Redundant

7+ Relay servers located distributed all over the world

2+ Access servers in Europe

provides Continuity of Service and High Performances



Scalable

Unlimited Runtimes and devices manageable

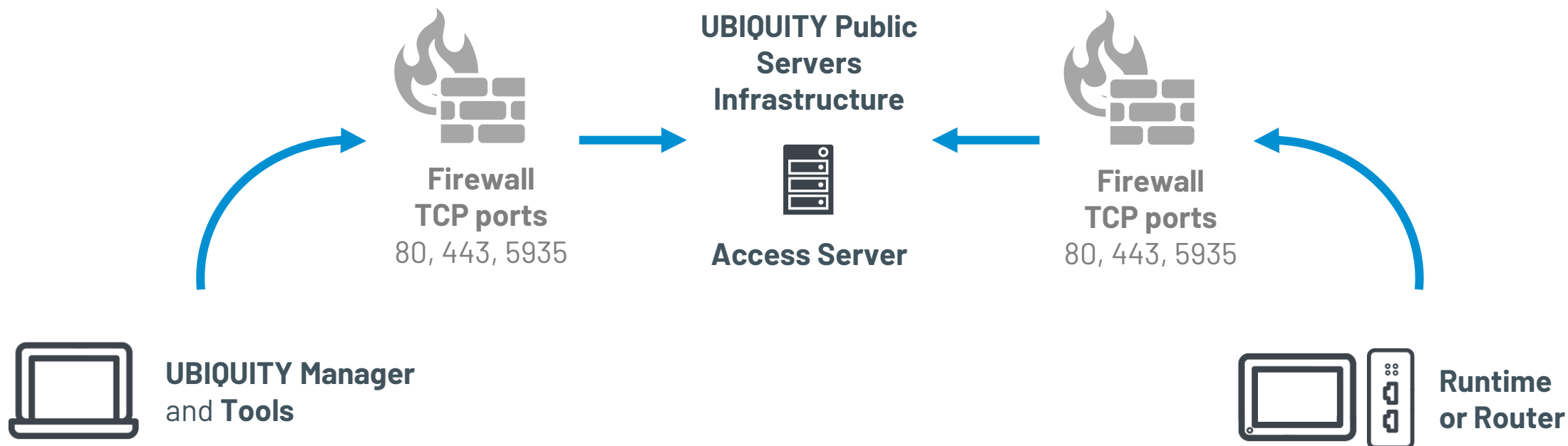
Unlimited Users and Groups

Unlimited traffic between UBIQUITY Manager and remote systems



OVERVIEW: PUBLIC SERVER INFRASTRUCTURE AUTHENTICATION

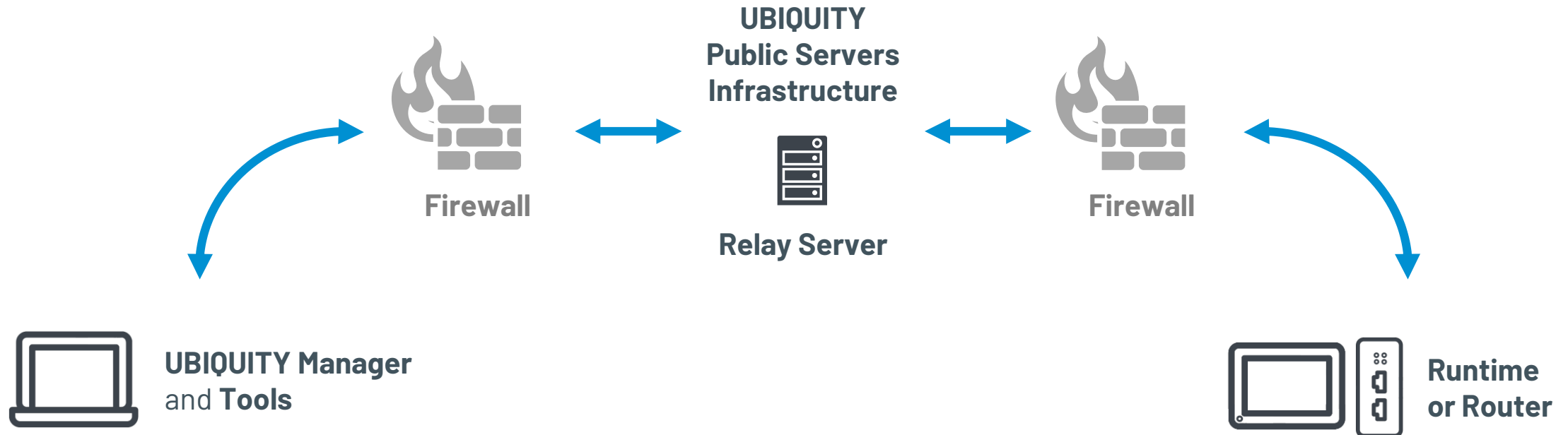
UBIQUITY Manager Tools, Runtime and Routers authenticates to one of the UBIQUITY Access Server using outgoing connections (SSL/TSL) usually allowed by firewall policies



OVERVIEW: PUBLIC SERVER INFRASTRUCTURE CONNECTION

When UBIQUITY Manager wants to connect to a Runtime/Router:

- Endpoints agree on the Relay Server with best round-trip time
- The secure end-to-end connection will be established with the Relay Server
- Relay Server forward encrypted messages without being able to decrypt them



UBIQUITY OVERVIEW: RUNTIME FOR IPCS



CROSS-PLATFORM APPLICATION

Installable on third-party iPC

pre-installed on all ASEM's iPC

Available from WinCE up to Win10
and Linux Ubuntu22



INTERACTIVE ACCESS

allows
interactive access
to remote systems
with audit



VPN ACCESS

Secure access
to the device
and to the
automation subnet

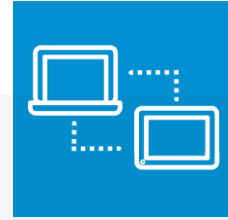
UBIQUITY OVERVIEW: HMIS



EMBEDDED OPERATOR INTERFACE

Ubiquity Runtime
Embedded

Can provide
physical network separation



INTERACTIVE ACCESS

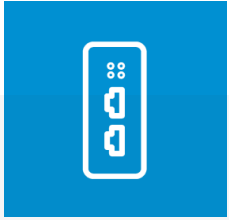
allows
interactive access
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VPN ACCESS

Secure access
to the device
and to the
automation subnet

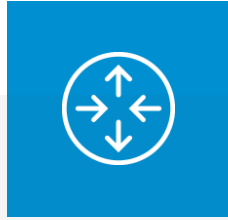
UBIQUITY OVERVIEW: ROUTERS



ROUTER APPLIANCE

Ubiquiti Runtime
embedded

Provide physical
network
separation



NETWORKING CAPABILITIES

Routing, NAT,
Internet Sharing
and can provide
Wifi/4G connection



EMBEDDED I/O

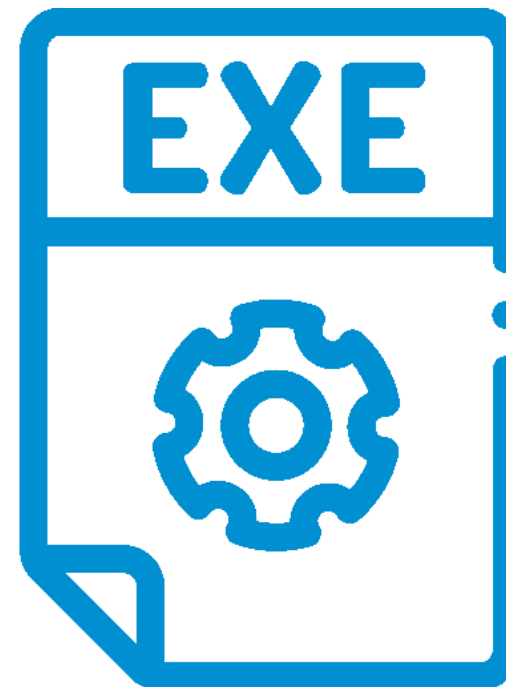
Remote access
management
via digital I/O



VPN ACCESS

Secure access
to the device
and to the
automation subnet

RUNTIME FEATURES



RUNTIMES FEATURES: INTERACTIVE ACCESS



REMOTE DESKTOP

VNC-like feature with
remote input lock and multi-monitor support

Allow to chat with remote users
with conversation log *

TASK MANAGER

Allow to check RAM and CPU usage
of process running on the remote system

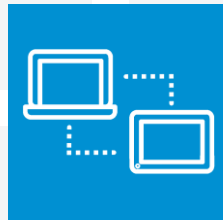
Process termination
Remote System Reboot

FILE TRANSFER

Allow to transfer files
to/from remote systems

MULTICLIENT SUPPORT

A remote system
manages multiple
concurrent connections



* chat is currently available only on Runtime for Windows and on ASEM HMI25/30/40

RUNTIMES FEATURES: VPN



VPN TO THE DEVICE

Optimized for industrial communications
The VPN Server is at Runtime level
not on the Server Infrastructure

IP CONFLICT RESOLUTION

Conflicts are solved
by UBIQUITY VPN virtual adapter
with the use of metric



DATA LINK - LAYER 2

No routing rules required
No default gateway required
Service PC gets a real remote subnet IP
Supports broadcast messages

FIREWALL and ROUTING RULES

VPN traffic can be filtered by Firewall rules
with a set of predefined rules ready-to-use
VPN traffic can be routed
by static routing rules

RUNTIMES FEATURES: ADVANCED



VPN TO THE SUBNET

VPN can be established with the automation's subnet to reach PLC, Drives of the machine/plant

INTERNET SHARING

Provide internet access to devices of automation's subnet
Filtered by MAC-Address



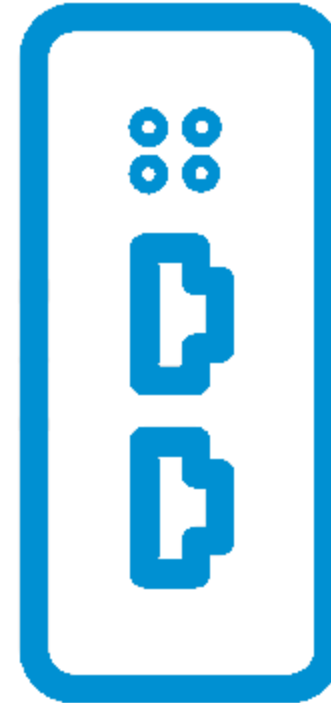
USB PASS-THROUGH

Allow to access a remote device connected to the remote system via USB port

SERIAL PASS-THROUGH

Allow to access a remote device connected to the remote system via Serial port

ROUTER FEATURES



UBIQUITY ROUTERS: WHY?

- if a **physical network separation** between the automation subnet and customer/internet subnet is required
- if a wired connection is not available, so a **Cellular or Wi-Fi connection** is needed
- if the **Runtime cannot be installed**, like operator panels with proprietary OS or managed by others



UBIQUITY ROUTERS: KEY FEATURES

NETWORKING CAPABILITIES

VPN and Advanced features
provided by the UBIQUITY Runtime
+
Routing and NAT

EASE OF MANAGEMENT

Configuration via USB Memory
or via System Manager

Remote firmware upgrade

PROTECTION

Protection against
unauthorized domain change

SECURITY INCREASE

Embedded I/O allows
to physically accept and check
the Remote Connection

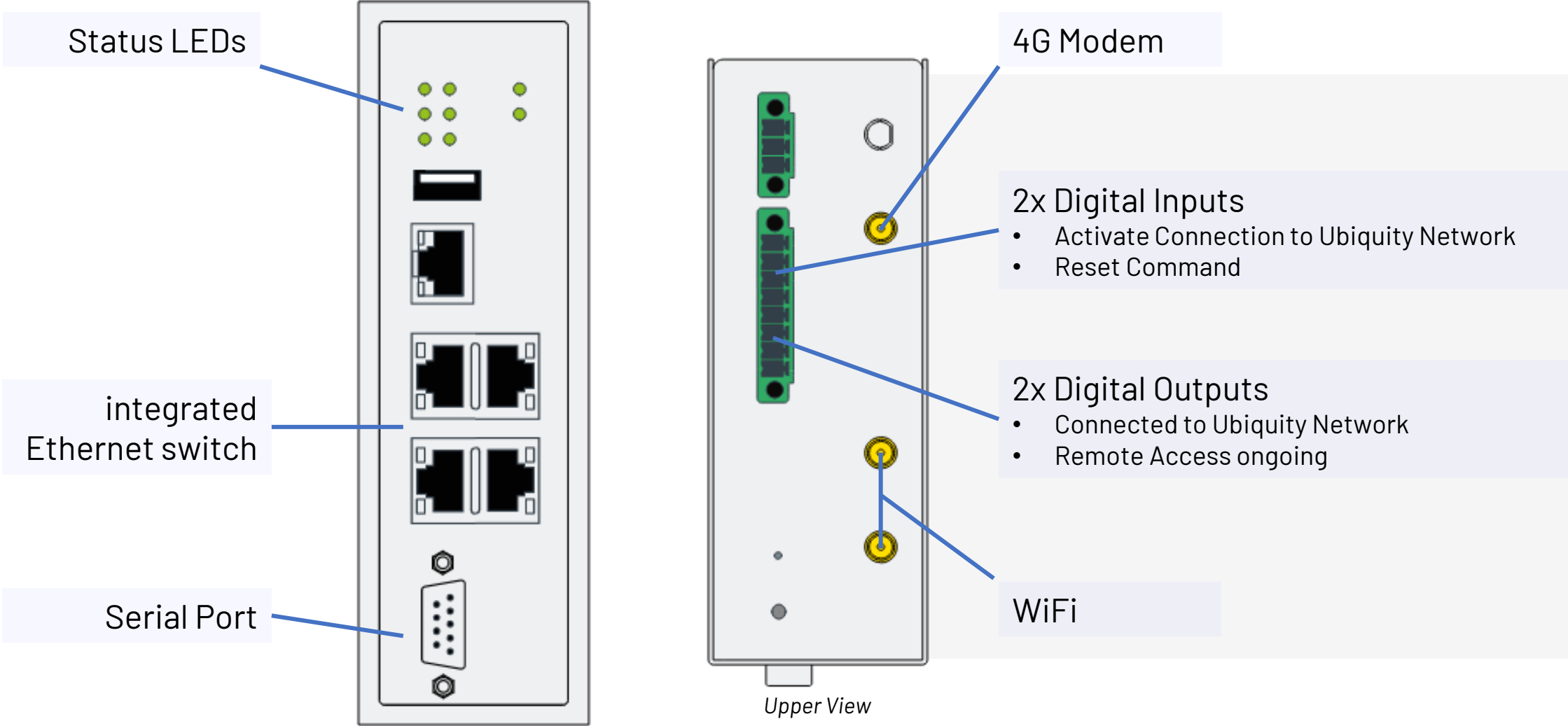


UBIQUITY ROUTERS

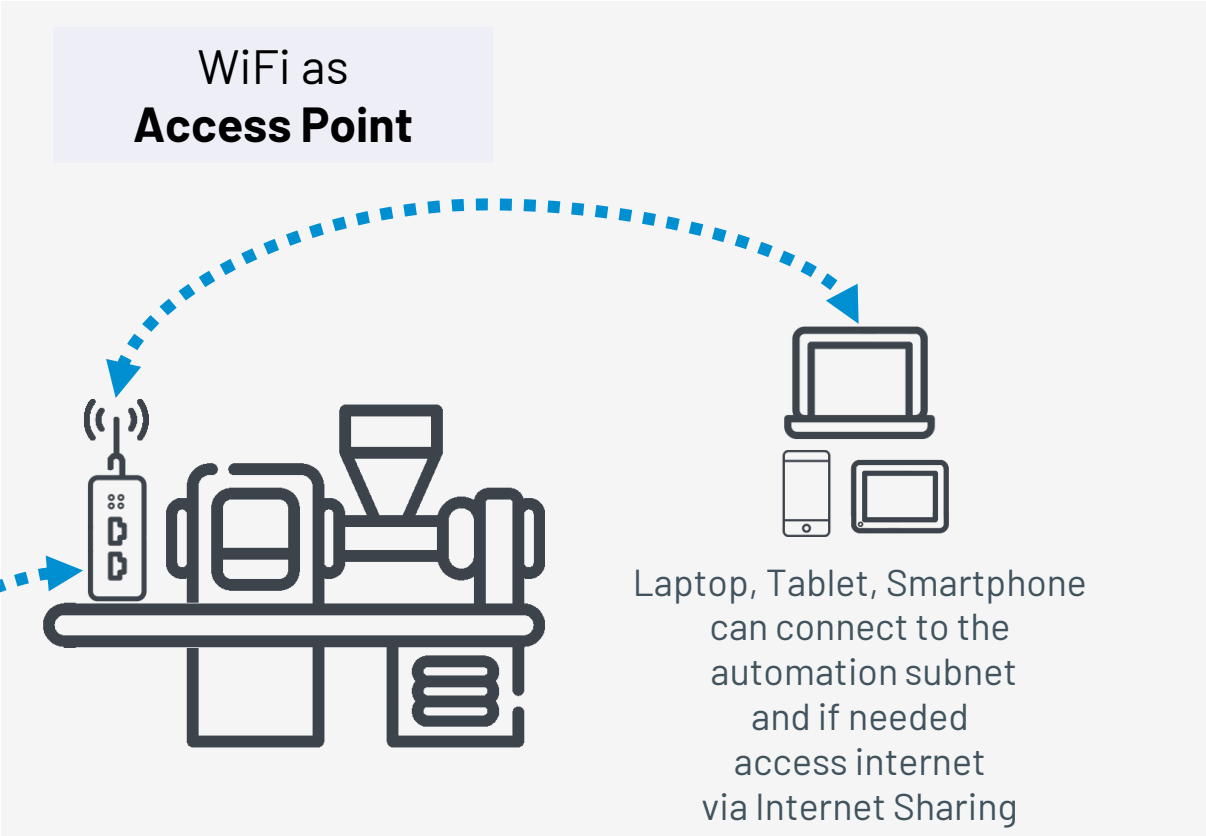
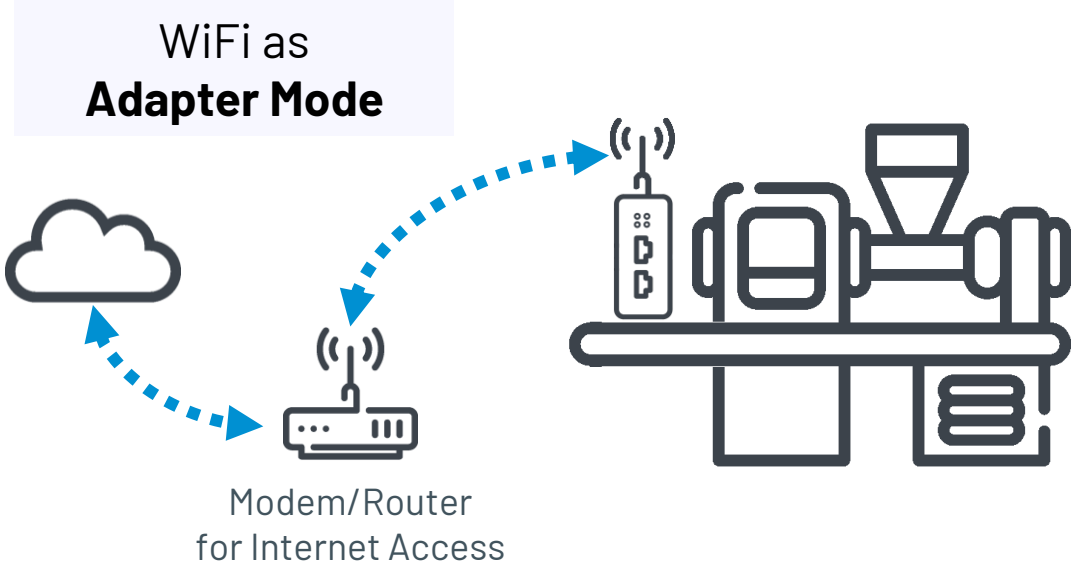


Features	RK20	RK21	RK22
USB 2.0	1	1	1
WAN Gigabit	1	1	1
LAN Gigabit	1	1	4
Serial port (multi-standard isolated)	✓	✓	✓
WiFi	X	✓	✓
4G Modem	X	✓	✓

UBIQUITY ROUTERS: DETAILED VIEW OF RK22



UBIQUITY ROUTERS: WIFI MODE



CONNECTIVITY SERVICES



CONNECTIVITY SERVICES

NEW
with 13.5

BEST ROUTE FOR REMOTE ACCESS

Automatic selection of the best route through one of the Relay Servers

LOCAL CONNECTION

Interactive Access to devices on the local network ⁽¹⁾



CONNECTION AUTHORIZATION

Operator/Technician can authorize the remote connection

OVER-THE-AIR UPDATES

Runtime/Firmware updates for remote devices

Updates can be scheduled

⁽¹⁾ works without Internet Access and without domain concurrent connection license
it's supported by the Windows Runtimes, OptixPanel HMIs and RK2x Routers (devices may require firmware update)

CONNECTIVITY SERVICES

NEW
with 14

DOMAIN MANAGEMENT

User profiling and
Permissions control

Subfolders act as sub-domains

DEVICE MANAGEMENT

Sub-devices definition
with actions/services

Invite user to a device

AUDIT

Audit of connections
and administration activities

GEOLOCATION

View of remote devices
on geographical map
based on public IP address



CONNECTIVITY SERVICES



APPS

Augmented Reality App
for video streaming with VoIP

VPN App

SINGLE SIGN-ON

Single Sign-On
interfacing with 3rd party
OpenID identity providers ⁽¹⁾



ASSISTANCE REQUEST

Operator/Technician
can send assistance request

Notifications are visible in UBIQUITY
Manager and/or via e-mail

3rd PARTY INTEGRATION

Integration
on 3rd party application
through Web API ⁽²⁾

⁽¹⁾ requires an Unlimited Concurrent Connections license and a feasibility assessment by ASEM Engineering

⁽²⁾ the web API is used for domain management, SDK allow to open VPN tunnel and File Transfer

LICENSE MODEL



LICENSING MODEL



Connectivity services

Based on the
**Number of concurrent
remote connections**
with devices

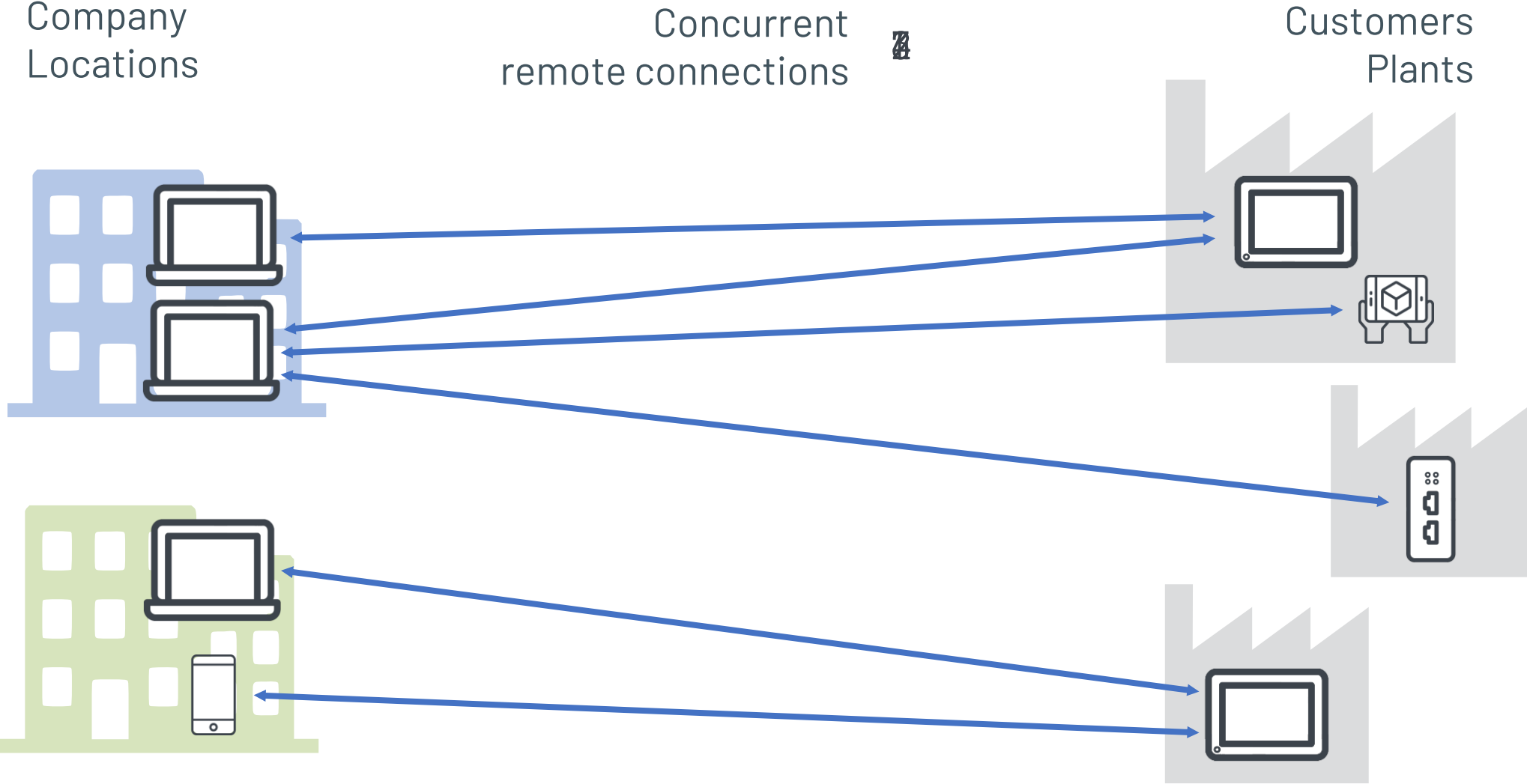


Runtimes

Available for
iPCs, HMIs and Routers

Perpetual License
Basic and **Pro**

CONNECTIVITY SERVICES: CONCURRENT CONNECTIONS



CONNECTIVITY SERVICES: DETAILS

UBIQUITY Connectivity Services

- Annual fee
- License based on an Activation Key to use on the Domain through UBIQUITY Manager
- Available sizes of concurrent remote connections
 - 1 remote connection
 - 2 concurrent remote connections
 - 5 concurrent remote connections
 - 10 concurrent remote connections
 - Unlimited concurrent remote connections ⁽¹⁾

⁽¹⁾ includes also the "Basic concurrent access" feature: any Runtime without a license will be recognized as with the Basic license

RUNTIMES LICENSES FOR IPC AND HMI



UBIQUITY Runtime for iPC ⁽¹⁾ and HMI ⁽²⁾	BASIC	PRO
▪ Interactive tools: Remote Desktop, Chat ⁽³⁾, File transfer, Task manager ▪ VPN to the device with integrated Firewall and Routing Rules ▪ Support for multiple connections from different UBIQUITY Manager Tools with separate VPNs for each client ▪ Local Connection to use Interactive tools and VPN without internet access	✓	✓
▪ VPN to the automation subnet with integrated Firewall and Routing Rules ▪ Serial and USB passthrough ▪ Internet connection sharing (ICS) with devices on the automation subnet	-	✓

⁽¹⁾ any ASEM iPC, includes Ubiquity Runtime Basic license
⁽²⁾ any ASEM HMI, includes Ubiquity Runtime Basic or Pro license depending on the model
⁽³⁾ available on Runtime for Windows and on ASEM HMI25/30/40

RUNTIMES LICENSES FOR ROUTERS



UBIQUITY Runtime for Router ⁽¹⁾	PRO ROUTER
▪ Interactive tools: File transfer, Task manager ▪ VPN to the device with integrated Firewall and Routing Rules ▪ Support for multiple connections from different UBIQUITY Manager Tools with separate VPNs for each client ▪ Local Connection to use Interactive tools and VPN without internet access	✓
▪ VPN to the automation subnet with integrated Firewall and Routing Rules ▪ Serial and USB passthrough ▪ Internet connection sharing (ICS) with devices on the automation subnet	✓
▪ Programming NAT rules between Ethernet interfaces ▪ Programming static Routing rules between Ethernet interfaces	✓

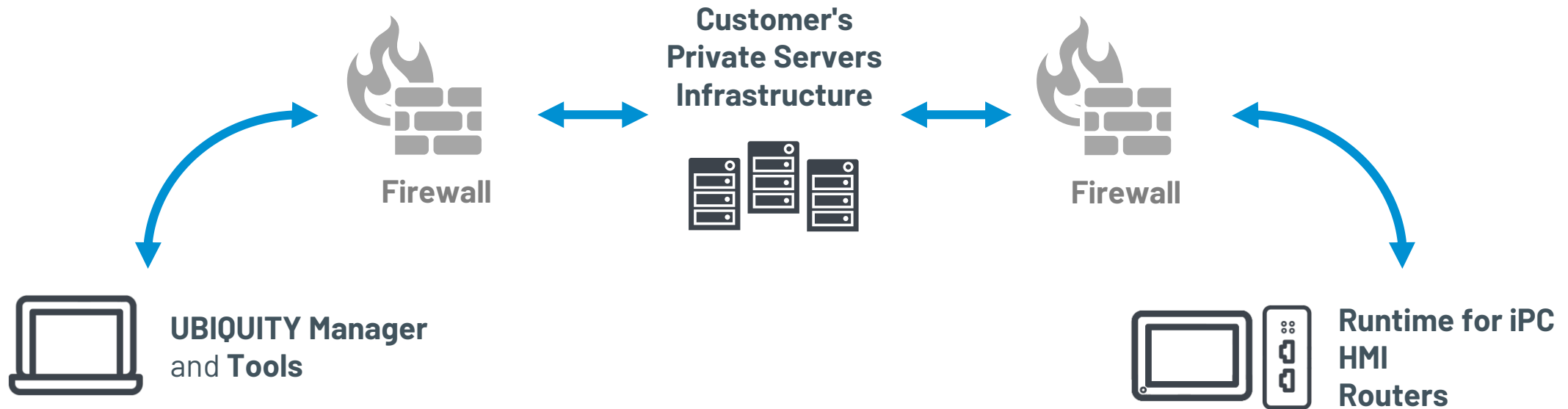
⁽¹⁾ any UBIQUITY Router includes the Ubiquity Pro Router license

PRIVATE SERVER



PRIVATE SERVER: OVERVIEW

- UBIQUITY Manager and UBIQUITY Manager Tools
- Ubiquity Private Server Infrastructure
- Runtime, HMI and Routers



PRIVATE SERVER: WHY?

FULL CONTROL

The IT department
can have full control
over the whole
servers infrastructure

Ubiquity servers can be installed
on physical or virtual machine
hosted on premises or on a private cloud



EXTRA SECURITY

Gather an extra degree of security
thanks to the separation
from the Public Infrastructure

Any possible data breach,
server maintenance,
connectivity downtime
on the public infrastructure
does not effect the private

PRIVATE SERVER: LICENSE MODEL



PRIMARY SERVER

Acts as Access Server
and Relay Server
with

Unlimited concurrent connections

Includes the first year
of Maintenance and Technical Support
Following years require annual payment.



SECONDARY SERVER

Acts only as a Relay Server

Can be used for redundancy,
decrease latency,
traffic balancing.

It's optional and require the presence
of the Primary Server but can be installed
in a distinct geographic location

PUBLIC VS PRIVATE SERVER

CONNECTIVITY SERVICES	PUBLIC	PRIVATE
▪ USER, GROUPS and SUB DOMAINS: User profiling and Permissions control	✓	
▪ AUDIT: Audit of connections and administration activities	✓	
▪ ASSISTANCE REQUEST and CONNECTION AUTHORIZATION: User can request assistance from the remote device and User can authorize the remote connection	✓	
▪ LOCAL CONNECTION: Interactive Access to devices on the local network	✓	
▪ SUB-DEVICES: Define devices in the automation subnetwork with easy access to their services	✓	
▪ INVITE USER TO DEVICE: Invite a user to access to a device with temporary permissions	✓	
▪ APPS: Augmented Reality App and VPN App (Android and iOS)	✓	
▪ BEST ROUTE : Automatic selection of the best route through one of the Relay Servers	✓	✓ ⁽¹⁾
▪ OVER-THE-AIR UPDATES: Runtime/Firmware updates for remote devices	✓	-
▪ MAP/GEOLOCATION: View of remote devices on geographical map based on public IP address	✓	✓ ⁽²⁾
▪ SINGLE SIGN-ON: Single Sign-On interfacing with 3rd party OpenID identity providers	✓ ⁽³⁾	✓
▪ 3rd PARTY INTEGRATION: Integration on 3rd party application through Web API and Tools SDK	✓	
▪ BASIC CONCURRENT ACCESS: any Runtime will be recognized as with the Basic license	✓ ⁽³⁾	✓

⁽¹⁾ require at least one Secondary Server license

⁽²⁾ require to purchase a Geolocation service from a 3rd party provider

⁽³⁾ available only with "Unlimited concurrent connections"

UBIQUITY VS FT REMOTE ACCESS





Part of FT HUB
Integration with FT Optix
Integration with FT Design Studio

Web Client
Runtime
for Windows10
and Linux Ubuntu22
Trial Period
License based on
Concurrent Connection
Unlimited License
Wired/Wireless Routers
Local Connection
Geolocation
Remote Device Update

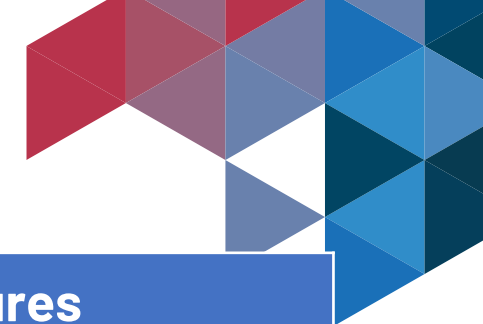
Device/Router Setup via USB ⁽¹⁾
Single Sign-On (with external OpenID Connect provider)
Runtime for WCE, WinXP, Win7
Augmented Reality App (includes VoIP)
VPN App
Private Server
Web API + SDK

⁽¹⁾ In the near future will be available also in FT Remote Access

UBIQUITY VS UBIQUITY X



UBIQUITY VS UBIQUITY X



UBIQUITY - Domain features ⁽¹⁾	UBIQUITY X - Domain features
✓ Best route selection, ✓ Geolocation, ✓ Remote Connection authorization, ✓ Remote Runtime/Firmware updates with scheduling, ✓ Users, Groups and permission management, ✓ Audit of Connections and Operations, ✓ WebAPI and SDK for 3rd party integration	✓ Best route selection, ✓ Geolocation, ✓ Remote Connection authorization, ✓ Remote Runtime/Firmware updates with scheduling, ✓ Users, Groups and permission management, ✓ Audit of Connections and Operations, ✓ WebAPI and SDK for 3rd party integration ✓ Access to new Relay Servers for low latency, ✓ Sub-devices, ✓ Invite user to a device, ✓ Assistance request, ✓ Single Sign-On with 3rd party OpenID identity providers ⁽²⁾ ✓ Augmented Reality App and VPN App for Android and iOS

⁽¹⁾ Features related to Runtimes and/or Embedded Devices remain available on both Ubiquity and Ubiquity X domains

⁽²⁾ Require license "Unlimited concurrent remote connections"

SECURITY ARCHITECTURE



SECURITY BY DESIGN: OUR FOUNDATIONAL PRINCIPLE



SECURITY FIRST APPROACH

We built Ubiquity with security as our cornerstone. This isn't an afterthought—it's embedded in our DNA.



STRATEGIC PRIORITIZATION

When making implementation decisions, security takes precedence. We never compromise protection for convenience.



SECURE CONNECTIONS ACROSS UNTRUSTED NETWORKS

Ubiquity's primary mission: establishing secure client connections to remote devices across the insecure Internet.

SECURE COMMUNICATIONS & ENCRYPTION



1

HTTPS COMMUNICATIONS

All communications with Access Servers and Web API utilize HTTPS on TCP port 443, ensuring encrypted data transmission. SSL certificates from trusted Certification Authorities authenticate server identities.

2

RELAY SERVER PROTECTION

Securely forward traffic between Clients and Runtime without requiring inbound ports, utilizing end-to-end AES-256 encryption with securely exchanged session keys.

3

SECURE PROTOCOLS

UBIQUITY Manager employs Secure Web Socket (WSS) protocol, providing an additional layer of protection for real time communication and control operations.

STRONG AUTHENTICATION & ACCESS MANAGEMENT



COMPLEX PASSWORDS

User authentication requires domain name, username and password, and domain admin can enforce complex passwords.



TWO-FACTOR AUTHENTICATION

Support for TOTP-based two-factor authentication (RFC 6238) compatible with applications like Google Authenticator provides an additional security layer beyond passwords.



AUTHORIZATION SYSTEM

The embedded authorization system allows administrators to define granular permissions for users, groups, and devices, ensuring principle of least privilege.

COMPREHENSIVE AUDIT & TRACEABILITY



COMPLETE AUDIT LOGGING

The system maintains a comprehensive audit log recording all operations including logins/logouts, configuration changes, and remote access activities.

TAMPER-PROOF RECORDS

Audit logs cannot be disabled or deleted, even by administrators, ensuring the integrity of security records.

REMOTE SESSION LOGGING

Administrators can enable detailed logging of all operations during remote access sessions, including transferred files and processes started or terminated, providing complete visibility into system interactions.

THREAT PROTECTION

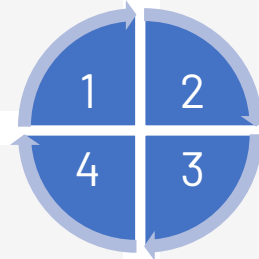


BRUTE FORCE PROTECTION

Countermeasures against brute force attacks include temporary IP address blocking after failed attempts.

SECURE PASSWORD STORAGE

Passwords are stored as salted hashes, protecting against rainbow table attacks and database breaches.



CERTIFIED CLOUD INFRASTRUCTURE

Hosted on a certified ISO 27001 cloud infrastructure, providing enterprise-grade infrastructure protection.

DIGITAL SIGNATURES

Binaries are digitally signed to guarantee authenticity and integrity, preventing tampering or malicious replacement.

VULNERABILITY MANAGEMENT & COMPLIANCE



VULNERABILITY MANAGEMENT

In Rockwell Automation, a dedicated Product Security Incident Response Team (PSIRT) handles product security incidents, while a 24x7 Security Operations Center monitors for threats.



SECURITY BY DESIGN IEC 62443

Our secure-development practices align with international standards IEC 62443 to help protect industrial systems and critical infrastructures.



SECURITY FRAMEWORKS

Rockwell Automation follows NIST Cybersecurity Framework, ensuring comprehensive security practices across identification, protection, detection, response, and recovery functions.



THANK YOU



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